Omotayo Harry Oludemi

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Education

McCormick School of Engineering, Northwestern University Evanston, IL
Bachelor of Science and Master of Science in Computer Science (Combined B.S./M.S. Program) Expected Jun 2027
Awards: QuestBridge Scholar; Peer Mentor Award, Northwestern CS
Relevant Coursework (Computer Science Fundamentals): Data Structures & Algorithms, Advanced Algorithms, System Design, Scalable Systems, Computer Security, Parallel Programming, Full-Stack Web Development, Human-Computer Interaction, Computer Vision

Skills

Languages & Scripting: Java, Python, JavaScript, TypeScript, HTML, CSS, C++, C
Server-Side Development: Spring Boot, REST APIs, GraphQL, Node.js, Express, Django, WebSockets, JPA, JDBC
Web Development (Frontend): React, Next.js, Redux Toolkit, Redux Saga, Material UI, Bootstrap
Databases: PostgreSQL, MySQL, Oracle Database, MongoDB, Redis, H2 Database
Cloud & DevOps: AWS, Docker, Kubernetes, GKP, Liquibase, Git, SonarQube
Testing & Tools: Unit and integration testing, React Testing Library, JIRA, MCPs

Work Experience

JPMorgan Chase & Co. Chicago, IL
Full Stack Software Engineer Intern Jun 2026 - Aug 2026

- Architected a real-time, event-driven server-side REST API in Java and Spring Boot for enterprise user onboarding, taking deployment cycles from 20 minutes to zero-downtime execution.
- Reduced maintenance overhead by 37% across distributed codebases by decomposing monolithic legacy logic into modular Spring Boot microservices, consolidating 8 repositories into 5.
- Migrated a runtime form engine to React and TypeScript at 100% functional parity, replacing dynamic configuration scripts with reusable components and cutting render latency by 20%.

CDK Global Hoffman Estates, IL
Software Development Intern Jun 2025 - Aug 2025

- Refactored legacy web front-end architecture into modular, responsive React and JavaScript components, improving rendering performance and maintainability for an enterprise application serving 9,528 clients.
- Reduced user scheduling friction by 30% by redesigning and implementing the CDK parts scheduler UI, achieving 100% compliance with core product requirements.
- Hardened platform stability ahead of release deadlines by running root-cause triage across 20+ Jira tickets, identifying critical defects, and shipping targeted fixes.

Computer Science Department, Northwestern University Evanston, IL
Teaching Assistant, Data Structures & Algorithms Jan 2025 - Present

- Mentor 100+ students per quarter in data structures and graph algorithms through weekly office hours, live debugging walk-throughs, and review sessions on topics such as Dijkstra's algorithm.
- Designed structured grading rubrics that standardized evaluation across 100+ students and delivered targeted feedback resolving recurring algorithmic misconceptions.
- Partnered with faculty across 3 quarters to author integration testing documentation and automated evaluation guidelines adopted department-wide.

Projects

FairPay | Java, Spring Boot, GraphQL, Next.js, WebSockets, Redis, PostgreSQL Evanston, IL

- Designed and deployed a high-throughput GraphQL API in Spring Boot with 9 mutations orchestrating shared expense calculations, user authentication, and real-time item claiming on PostgreSQL.
- Reduced API response latency by 50% under high concurrency by moving Google Gemini OCR receipt parsing off the synchronous request path into a Redis background worker queue.
- Eliminated client-side polling with STOMP-based WebSocket channels that broadcast real-time state changes to 20+ concurrent users per room at sub-350 ms latency.

Leadership Experience

Management Leadership for Tomorrow Feb 2025 - Present
Career Prep Fellow

- Selected for an 18-month professional development program for high-achieving diverse talent, building leadership and communication skills through Fortune 500 partner engagements and recruiting events.